Pérdida por Fricción - Tubería Plástica PVC Céd. 80 IPS en Pies por cada 100 pies de tubería

Calibre Tubo	1/2"		3/4"		1"		1 1/4"		1 1/2"		2"		2 1/2"		3"		4"	
GPM	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)	Pérd. Carga	Vel (ft./s)
1	1.87	1.37	0.42	0.74	0.12	0.45	0.03	0.25	0.01	0.18								
2	6.74	2.74	1.52	1.48	0.44	0.89	0.11	0.50	0.05	0.36	0.01	0.22						
3	14.29	4.11	3.21	2.23	0.93	1.34	0.23	0.75	0.10	0.54	0.03	0.33	0.01	0.23				
4	24.34	5.48	5.47	2.97	1.59	1.78	0.39	1.00	0.18	0.73	0.05	0.43	0.02	0.30				
5	36.79	6.85	8.27	3.71	2.40	2.23	0.59	1.25	0.27	0.91	0.08	0.54	0.03	0.38	0.01	0.24		
6	51.57	8.22	11.59	4.45	3.36	2.68	0.82	1.50	0.38	1.09	0.11	0.65	0.04	0.45	0.02	0.29		
7	68.61	9.59	15.43	5.19	4.47	3.12	1.09	1.75	0.50	1.27	0.14	0.76	0.06	0.53	0.02	0.34		
8	87.86	10.96	19.75	5.94	5.73	3.57	1.40	2.00	0.64	1.45	0.18	0.87	0.08	0.61	0.03	0.39		
9	109.27	12.33	24.57	6.68	7.12	4.01	1.74	2.25	0.80	1.63	0.23	0.98	0.01	0.68	0.03	0.44		
10	132.82	13.70	29.86	7.42	8.66	4.46	2.12	2.50	0.97	1.82	0.28	1.09	0.12	0.76	0.04	0.49		
11			35.63	8.16	10.33	4.91	2.53	2.75	1.16	2.00	0.33	1.20	0.14	0.83	0.05	0.53		
12			41.86	8.90	12.14	5.35	2.97	3.00	1.36	2.18	0.39	1.30	0.16	0.91	0.06	0.58		
14			55.69	10.39	16.15	6.24	3.95	3.50	1.81	2.54	0.52	1.52	0.22	1.06	0.07	0.68		
16			71.31	11.87	20.68	7.14	5.06	4.00	2.32	2.90	0.67	1.74	0.28	1.21	0.09	0.78		
18			88.69	13.36	25.72	8.03	6.30	4.50	2.89	3.27	0.83	1.96	0.34	1.36	0.12	0.87		
20			107.80	14.84	31.26	8.92	7.65	5.00	3.51	3.63	1.01	2.17	0.42	1.51	0.14	0.97	0.04	0.56
22					37.29	9.81	9.13	5.50	4.19	3.99	1.20	2.39	0.50	1.67	0.17	1.07	0.04	0.61
24					43.81	10.70	10.72	6.00	4.92	4.36	1.41	2.61	0.59	1.82	0.20	1.17	0.05	0.67
26					50.81	11.60	12.44	6.50	5.71	4.72	1.64	2.83	0.68	1.97	0.23	1.26	0.06	0.73
28					58.29	12.49	14.27	7.00	6.55	5.08	1.88	3.04	0.78	2.12	0.26	1.36	0.07	0.78
30					66.23	13.38	16.21	7.50	7.44	5.45	2.13	3.26	0.89	2.27	0.30	1.46	0.08	0.84
35							21.57	8.75	9.89	6.35	2.84	3.80	1.18	2.65	0.40	1.70	0.10	0.98
40							27.62	10.00	12.67	7.26	3.63	4.35	1.51	3.03	0.51	1.94	0.13	1.12
45	**Tubo 6"**						34.36	11.25	15.76	8.17	4.52	4.89	1.88	3.41	0.64	2.19	0.17	1.26
50	0.03	0.62					41.76	12.51	19.16	9.08	5.49	5.44	2.28	3.78	0.77	2.43	0.20	1.40
55	0.03	0.68					49.82	13.76	22.85	9.99	6.55	5.98	2.72	4.16	0.92	2.67	0.24	1.53
60	0.04	0.74					58.53	15.01	26.85	10.89	7.70	6.52	3.20	4.54	1.09	2.91	0.28	1.67
65	0.04	0.80							31.14	11.80	8.93	7.07	3.71	4.92	1.26	3.16	0.33	1.81
70	0.05	0.86							35.72	12.71	10.24	7.61	4.25	5.30	1.45	3.40	0.38	1.95
75	0.06	0.92							40.59	13.62	11.64	8.15	4.83	5.68	1.64	3.64	0.43	2.09
80	0.07	0.98							45.74	14.52	13.12	8.70	5.45	6.06	1.85	3.89	0.48	2.23
85	0.07	1.05							51.18	15.43	14.68	9.24	6.09	6.43	2.07	4.13	0.54	2.37
90	0.08	1.11							56.89	16.34	16.32	9.78	6.77	6.81	2.30	4.37	0.60	2.51
95	0.09	1.17							62.88	17.25	18.03	10.33	7.49	7.19	2.54	4.61	0.66	2.65
100	0.01	1.23	**Tubo 8"**						69.15	18.16	19.83	10.87	8.23	7.57	2.80	4.86	0.73	2.79
125	0.15	1.54	0.04	0.88							29.98	13.59	12.45	9.46	4.23	6.07	1.01	3.49
150	0.21	1.85	0.05	1.05							42.02	16.31	17.45	11.35	5.93	7.29	1.54	4.19
175	0.28	2.15	0.07	1.23	**Tubo 10"**								23.21	13.25	7.89	8.50	2.05	4.88
200	0.36	2.46	0.09	1.41	0.03	0.89							29.72	15.14	10.01	9.71	2.62	5.58
250	0.54	3.08	0.14	1.76	0.05	1.12							44.93	18.92	15.27	12.14	3.96	6.98
300	0.76	3.69	0.19	2.11	0.06	1.34	**Tubo 12"**										5.56	8.37
350	1.01	4.31	0.26	2.46	0.09	1.56	0.04	1.11									7.39	9.77
400	1.29	4.92	0.33	2.81	0.11	1.79	0.05	1.26									9.47	11.16
450	1.61	5.54	0.41	3.16	0.14	2.01	0.06	1.42										
500	1.95	6.15	0.50	3.51	0.17	2.23	0.07	1.58										
750	4.14	9.23	1.06	5.27	0.35	3.35	0.15	2.37										
1000	7.05	12.31	1.80	7.03	0.60	4.47	0.26	3.16										
1250			2.73	8.78	0.91	5.58	0.39	3.95										
1500			3.82	10.54	1.27	6.70	0.55	4.74										
2000					2.16	8.94	0.93	6.32										
2500					3.27	11.17	1.41	7.89										
3000					4.58	13.40	1.97	9.47										
3500							2.62	11.05										
4000							3.36	12.63										

Nota: Las velocidades en la succión no deben exceder los 8 pies por segundo. Calculado usando la fórmula de Williams y Hazen con C = 150.

Pérdida por Fricción - Agua en Pies por 100 pies de Tubo de Acero

GPM	Calibre Tubo 1/2" Calibre real 0.622			Calibre Tubo 3/4" Calibre real 0.824			Calibre Tubo 1" Calibre real 1.049			Calibre Tubo 1-1/4" Calibre real 1.38			Calibre Tubo 1-1/2" Calibre real 1.61		
	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)
1	2.099	1.056	0.017	0.534	0.602	0.006	0.165	0.371	0.002	0.043	0.215	0.001	0.021	0.158	0.000
2	7.577	2.112	0.069	1.929	1.203	0.023	0.596	0.742	0.009	0.157	0.429	0.003	0.074	0.315	0.002
3	16.056	3.168	0.156	4.087	1.805	0.051	1.263	1.114	0.019	0.332	0.644	0.006	0.157	0.473	0.003
4	27.354	4.223	0.277	6.962	2.407	0.090	2.151	1.485	0.034	0.566	0.858	0.011	0.268	0.630	0.006
5	41.352	5.279	0.433	10.525	3.008	0.141	3.252	1.856	0.054	0.856	1.073	0.018	0.404	0.788	0.001
6	57.961	6.335	0.624	14.753	3.610	0.203	4.558	2.227	0.077	1.200	1.287	0.026	0.567	0.946	0.014
7	77.112	7.391	0.849	19.628	4.211	0.276	6.064	2.599	0.105	1.597	1.502	0.035	0.754	1.103	0.019
8	98.747	8.447	1.109	25.134	4.813	0.360	7.765	2.970	0.137	2.045	1.716	0.046	0.966	1.261	0.025
9	122.817	9.503	1.404	31.261	5.415	0.456	9.657	3.341	0.173	2.543	1.931	0.058	1.201	1.418	0.031
10	149.280	10.559	1.733	37.997	6.016	0.563	11.738	3.712	0.214	3.091	2.145	0.072	1.460	1.576	0.039
11	178.099	11.615	2.097	45.332	6.618	0.681	14.004	4.083	0.259	3.688	2.360	0.087	1.742	1.734	0.047
12	209.241	12.670	2.495	53.259	7.220	0.810	16.453	4.455	0.308	4.333	2.574	0.103	2.047	1.891	0.056
13	242.675	13.726	2.928	61.769	7.821	0.951	19.082	4.826	0.362	5.025	2.789	0.121	2.374	2.049	0.065
14	278.376	14.782	3.396	70.856	8.423	1.103	21.889	5.197	0.420	5.764	3.003	0.140	2.723	2.206	0.076
15	316.317	15.838	3.899	80.513	9.025	1.266	24.873	5.568	0.482	6.550	3.218	0.161	3.094	2.364	0.087
16				90.735	9.626	1.440	28.031	5.940	0.548	7.381	3.432	0.183	3.487	2.521	0.099
17				101.517	10.228	1.626	31.362	6.311	0.619	8.258	3.647	0.207	3.901	2.679	0.112
18				112.852	10.829	1.823	34.863	6.682	0.694	9.181	3.861	0.232	4.337	2.837	0.125
19				124.738	11.431	2.031	38.535	7.053	0.773	10.148	4.076	0.258	4.793	2.994	0.139
20				137.168	12.033	2.250	42.375	7.425	0.857	11.159	4.290	0.286	5.271	3.152	0.154
21				150.140	12.634	2.481	46.383	7.796	0.945	12.214	4.505	0.315	5.769	3.309	0.170
22				163.649	13.236	2.723	50.556	8.167	1.037	13.313	4.719	0.346	6.288	3.467	0.187
23				177.691	13.838	2.976	54.894	8.538	1.133	14.455	4.934	0.378	6.828	3.625	0.204
24				192.263	14.439	3.241	59.396	8.909	1.234	15.641	5.148	0.412	7.388	3.782	0.222
25				207.363	15.041	3.516	64.061	9.281	1.339	16.869	5.363	0.447	7.968	3.940	0.241
26				222.985	15.643	3.803	68.887	9.652	1.448	18.140	5.577	0.483	8.569	4.097	0.261
27				239.128	16.244	4.101	73.874	10.023	1.561	19.453	5.792	0.521	9.189	4.255	0.281
28				255.789	16.846	4.411	79.021	10.394	1.679	20.809	6.006	0.561	9.829	4.413	0.303
29				272.965	17.448	4.731	84.327	10.766	1.801	22.206	6.221	0.601	10.489	4.570	0.325
30				290.652	18.049	5.063	89.791	11.137	1.928	23.645	6.435	0.644	11.169	4.728	0.347
32							101.191	11.879	2.193	26.647	6.864	0.732	12.587	5.043	0.395
34							113.215	12.622	2.476	29.813	7.293	0.827	14.082	5.358	0.446
36							125.857	13.364	2.776	33.142	7.722	0.927	15.655	5.673	0.500
38							139.112	14.107	3.093	36.633	8.151	1.033	17.304	5.989	0.557
40							152.975	14.849	3.427	40.283	8.580	1.144	19.028	6.304	0.618
42							167.442	15.592	3.778	44.093	9.009	1.261	20.827	6.619	0.681
44							182.507	16.334	4.147	48.060	9.438	1.384	22.701	6.934	0.747
46							198.168	17.076	4.532	52.184	9.867	1.513	24.649	7.249	0.817
48							214.420	17.819	4.935	56.463	10.296	1.648	26.671	7.564	0.889
50							231.259	18.561	5.355	60.898	10.725	1.788	28.765	7.880	0.965
55							275.904	20.417	6.479	72.654	11.798	2.163	34.319	8.668	1.168
60							324.147	22.274	7.711	85.358	12.870	2.574	40.319	9.456	1.390
65							375.943	24.130	9.049	98.997	13.943	3.021	46.762	10.244	1.631
70							431.249	25.986	10.495	113.561	15.015	3.504	53.641	11.032	1.891
75							490.027	27.842	12.048	129.039	16.088	4.023	60.952	11.820	2.171
80							552.241	29.698	13.708	145.422	17.160	4.577	68.691	12.607	2.470
85							617.860	31.554	15.475	162.701	18.233	5.167	76.853	13.395	2.789
90							686.852	33.410	17.349	180.869	19.305	5.793	85.435	14.183	3.127
95							759.190	35.267	19.331	199.918	20.378	6.454	94.433	14.971	3.484
100							834.846	37.123	21.419	219.840	21.450	7.151	103.843	15.759	3.860
110										262.281	23.595	8.653	123.890	17.335	4.671
120										308.142	25.740	10.298	145.553	18.911	5.559
130										357.380	27.885	12.086	168.811	20.487	6.524
140										409.955	30.030	14.016	193.645	22.063	7.566
150										465.831	32.175	16.090	220.039	23.639	8.685
160													247.975	25.215	9.882
170													277.440	26.791	11.156
180													308.420	28.367	12.507
190													340.902	29.943	13.935
200													374.874	31.519	15.440

Nota: Las velocidades en la succión no deben exceder los 8 pies por segundo.

Pérdida por Fricción - Agua en Pies por 100 pies de Tubo de Acero

GPM	Calibre Tubo 2"			Calibre Tubo 2-1/2"			Calibre Tubo 3"			Calibre Tubo 3-1/2"		
	Calibre real 2.067			Calibre real 2.469			Calibre real 3.068			Calibre real 3.548		
	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)
2	0.022	0.191	0.001									
3	0.047	0.287	0.001	0.020	0.201	0.001						
4	0.079	0.383	0.002	0.033	0.268	0.001						
5	0.120	0.478	0.004	0.051	0.335	0.002	0.018	0.217	0.001	0.009	0.162	0.000
6	0.168	0.574	0.005	0.071	0.402	0.003	0.025	0.260	0.001	0.012	0.195	0.001
7	0.224	0.670	0.007	0.094	0.469	0.003	0.033	0.304	0.001	0.016	0.227	0.001
8	0.286	0.765	0.009	0.121	0.536	0.004	0.042	0.347	0.002	0.021	0.260	0.001
9	0.356	0.861	0.012	0.150	0.603	0.006	0.052	0.391	0.002	0.026	0.292	0.001
10	0.433	0.957	0.014	0.182	0.670	0.007	0.063	0.434	0.003	0.031	0.325	0.002
11	0.516	1.052	0.017	0.218	0.737	0.008	0.076	0.477	0.004	0.037	0.357	0.002
12	0.607	1.148	0.020	0.256	0.804	0.010	0.089	0.521	0.004	0.044	0.389	0.002
13	0.704	1.244	0.024	0.296	0.871	0.012	0.103	0.564	0.005	0.051	0.422	0.003
14	0.807	1.339	0.028	0.340	0.938	0.014	0.118	0.608	0.006	0.058	0.454	0.003
15	0.917	1.435	0.032	0.386	1.005	0.016	0.134	0.651	0.007	0.066	0.487	0.004
16	1.034	1.531	0.036	0.435	1.072	0.018	0.151	0.694	0.007	0.075	0.519	0.004
17	1.157	1.626	0.041	0.487	1.139	0.020	0.169	0.738	0.008	0.083	0.552	0.005
18	1.286	1.722	0.046	0.542	1.206	0.023	0.188	0.781	0.009	0.093	0.584	0.005
19	1.421	1.818	0.051	0.599	1.273	0.025	0.208	0.825	0.011	0.103	0.617	0.006
20	1.563	1.913	0.057	0.658	1.340	0.028	0.229	0.868	0.012	0.113	0.649	0.007
21	1.711	2.009	0.063	0.720	1.407	0.031	0.250	0.911	0.013	0.123	0.681	0.007
22	1.865	2.105	0.069	0.785	1.474	0.034	0.273	0.955	0.014	0.135	0.714	0.008
23	2.025	2.200	0.075	0.853	1.541	0.037	0.296	0.998	0.015	0.146	0.746	0.009
24	2.191	2.296	0.082	0.923	1.608	0.040	0.321	1.042	0.017	0.158	0.779	0.009
25	2.363	2.391	0.089	0.995	1.675	0.044	0.346	1.085	0.018	0.171	0.811	0.010
26	2.541	2.487	0.096	1.070	1.742	0.047	0.372	1.128	0.020	0.183	0.844	0.011
27	2.724	2.583	0.104	1.148	1.809	0.051	0.399	1.172	0.021	0.197	0.876	0.012
28	2.914	2.678	0.112	1.227	1.876	0.055	0.427	1.215	0.023	0.210	0.909	0.013
29	3.110	2.774	0.120	1.310	1.943	0.059	0.455	1.259	0.025	0.224	0.941	0.014
30	3.312	2.870	0.128	1.395	2.010	0.063	0.485	1.302	0.026	0.239	0.974	0.015
32	3.732	3.061	0.146	1.572	2.144	0.071	0.546	1.389	0.030	0.269	1.038	0.017
34	4.175	3.252	0.164	1.759	2.278	0.081	0.611	1.476	0.034	0.301	1.103	0.019
36	4.642	3.444	0.184	1.955	2.412	0.090	0.679	1.562	0.038	0.335	1.168	0.021
38	5.130	3.635	0.205	2.161	2.546	0.101	0.751	1.649	0.042	0.370	1.233	0.024
40	5.642	3.826	0.228	2.376	2.680	0.112	0.826	1.736	0.047	0.407	1.298	0.026
42	6.175	4.018	0.251	2.601	2.814	0.123	0.904	1.823	0.052	0.446	1.363	0.029
44	6.731	4.209	0.275	2.835	2.948	0.135	0.985	1.910	0.057	0.486	1.428	0.032
46	7.308	4.400	0.301	3.078	3.083	0.148	1.070	1.996	0.062	0.527	1.493	0.035
48	7.908	4.592	0.328	3.331	3.217	0.161	1.158	2.083	0.067	0.571	1.558	0.038
50	8.529	4.783	0.356	3.592	3.351	0.174	1.248	2.170	0.073	0.616	1.623	0.041
55	10.175	5.261	0.430	4.286	3.686	0.211	1.490	2.387	0.089	0.734	1.785	0.050
60	11.955	5.740	0.512	5.035	4.021	0.251	1.750	2.604	0.105	0.863	1.947	0.059
65	13.865	6.218	0.601	5.840	4.356	0.295	2.030	2.821	0.124	1.001	2.109	0.069
70	15.905	6.696	0.697	6.699	4.691	0.342	2.328	3.038	0.143	1.148	2.272	0.080
75	18.072	7.174	0.800	7.612	5.026	0.393	2.646	3.255	0.165	1.304	2.434	0.092
80	20.367	7.653	0.910	8.578	5.361	0.447	2.981	3.472	0.187	1.470	2.596	0.105
85	22.787	8.131	1.028	9.598	5.696	0.504	3.336	3.689	0.212	1.644	2.758	0.118
90	25.331	8.609	1.152	10.669	6.031	0.565	3.708	3.906	0.237	1.828	2.921	0.133
95	27.999	9.088	1.284	11.793	6.366	0.630	4.099	4.123	0.264	2.021	3.083	0.148
100	30.789	9.566	1.422	12.968	6.701	0.698	4.507	4.340	0.293	2.222	3.245	0.164
110	36.733	10.523	1.721	15.472	7.371	0.845	5.377	4.774	0.354	2.651	3.570	0.198
120	43.156	11.479	2.048	18.177	8.041	1.005	6.317	5.208	0.422	3.114	3.894	0.236
130	50.052	12.436	2.404	21.081	8.711	1.180	7.327	5.642	0.495	3.612	4.219	0.277
140	57.416	13.392	2.788	24.183	9.382	1.368	8.405	6.076	0.574	4.144	4.543	0.321
150	65.241	14.349	3.200	27.479	10.052	1.570	9.550	6.510	0.659	4.708	4.868	0.368
160	73.524	15.306	3.641	30.968	10.722	1.787	10.763	6.944	0.749	5.306	5.192	0.419
170	82.261	16.262	4.110	34.647	11.392	2.017	12.042	7.378	0.846	5.937	5.517	0.473
180	91.446	17.219	4.608	38.516	12.062	2.261	13.386	7.812	0.948	6.599	5.841	0.530
190	101.077	18.175	5.134	42.573	12.732	2.520	14.796	8.246	1.057	7.295	6.166	0.591
200	111.150	19.132	5.689	46.815	13.402	2.792	16.271	8.680	1.171	8.021	6.490	0.655
220	132.607	21.045	6.884	55.853	14.742	3.378	19.412	9.548	1.417	9.570	7.139	0.792
240	155.794	22.958	8.192	65.619	16.083	4.020	22.806	10.416	1.686	11.243	7.788	0.943
260	180.689	24.872	9.614	76.104	17.423	4.718	26.450	11.284	1.979	13.040	8.437	1.106
280	207.270	26.785	11.150	87.300	18.763	5.472	30.341	12.152	2.295	14.958	9.086	1.283
300	235.521	28.698	12.800	99.199	20.103	6.281	34.476	13.020	2.635	16.997	9.735	1.473
350	313.339	33.481	17.423	131.975	23.454	8.550	45.868	15.190	3.586	22.613	11.358	2.005
400	401.250	38.264	22.756	169.002	26.805	11.167	58.737	17.360	4.684	28.957	12.980	2.619
450				210.197	30.155	14.133	73.054	19.530	5.928	36.016	14.603	3.314
500				255.488	33.506	17.448	88.795	21.699	7.318	43.776	16.225	4.092
550				304.810	36.856	21.113	105.937	23.869	8.855	52.227	17.848	4.951
600				358.108	40.207	25.126	124.460	26.039	10.539	61.359	19.470	5.892
650							144.348	28.209	12.368	71.164	21.093	6.915
700							165.583	30.379	14.344	81.633	22.715	8.020
750							188.152	32.549	16.466	92.759	24.338	9.206
800							212.040	34.719	18.735	104.536	25.960	10.475
850							237.235	36.889	21.150	116.957	27.583	11.825
900							263.725	39.059	23.712	130.017	29.206	13.257
950							291.500	41.229	26.420	143.710	30.828	14.771
1000							320.549	43.399	29.274	158.032	32.451	16.367
1100										188.540	35.696	19.804
1200										221.507	38.941	23.568
1300										256.902	42.186	27.660
1400										294.695	45.431	32.079
1500										334.861	48.676	36.825

Nota: Las velocidades en la succión no deben exceder los 8 pies por segundo.

Pérdida por Fricción - Agua en Pies por 100 pies de Tubo de Acero

GPM	Calibre Tubo 4" Calibre real 4.026			Calibre Tubo 5" Calibre real 5.047			Calibre Tubo 6" Calibre real 6.065		
	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)	Fricción	Vel (ft/seg)	Carga Vel (ft)
20	0.061	0.504	0.004						
21	0.067	0.529	0.004						
22	0.073	0.554	0.005						
23	0.079	0.580	0.005						
24	0.085	0.605	0.006						
25	0.092	0.630	0.006						
26	0.099	0.655	0.007						
27	0.106	0.680	0.007						
28	0.114	0.706	0.008						
29	0.121	0.731	0.008						
30	0.129	0.756	0.009						
32	0.146	0.806	0.010						
34	0.163	0.857	0.011						
36	0.181	0.907	0.013						
38	0.200	0.958	0.014						
40	0.220	1.008	0.016	0.073	0.641	0.006			
42	0.241	1.059	0.017	0.080	0.674	0.007			
44	0.263	1.109	0.019	0.087	0.706	0.008			
46	0.285	1.159	0.021	0.095	0.738	0.008			
48	0.309	1.210	0.023	0.103	0.770	0.009			
50	0.333	1.260	0.025	0.111	0.802	0.001			
55	0.397	1.386	0.030	0.132	0.882	0.012			
60	0.466	1.512	0.036	0.155	0.962	0.014			
65	0.541	1.638	0.042	0.180	1.042	0.017			
70	0.621	1.764	0.048	0.207	1.123	0.020			
75	0.705	1.890	0.056	0.235	1.203	0.022			
80	0.795	2.016	0.063	0.265	1.283	0.026			
85	0.889	2.142	0.071	0.296	1.363	0.029			
90	0.988	2.268	0.080	0.329	1.443	0.032			
95	1.092	2.394	0.089	0.364	1.524	0.036			
100	1.201	2.520	0.099	0.400	1.604	0.040	0.164	1.111	0.019
110	1.433	2.772	0.119	0.477	1.764	0.048	0.195	1.222	0.023
120	1.684	3.024	0.142	0.561	1.924	0.058	0.229	1.333	0.028
130	1.953	3.276	0.167	0.650	2.085	0.068	0.266	1.444	0.032
140	2.240	3.528	0.193	0.746	2.245	0.078	0.305	1.555	0.038
150	2.546	3.780	0.222	0.848	2.406	0.090	0.347	1.666	0.043
160	2.869	4.032	0.253	0.955	2.566	0.102	0.391	1.777	0.049
170	3.210	4.284	0.285	1.069	2.726	0.116	0.437	1.888	0.055
180	3.568	4.536	0.320	1.188	2.887	0.130	0.486	1.999	0.062
190	3.944	4.788	0.356	1.313	3.047	0.144	0.537	2.110	0.069
200	4.337	5.040	0.395	1.444	3.207	0.160	0.591	2.221	0.077
220	5.174	5.545	0.478	1.723	3.528	0.193	0.705	2.443	0.093
240	6.079	6.049	0.569	2.024	3.849	0.230	0.828	2.665	0.110
260	7.050	6.553	0.667	2.348	4.170	0.270	0.960	2.887	0.130
280	8.087	7.057	0.774	2.693	4.490	0.313	1.101	3.109	0.150
300	9.190	7.561	0.888	3.060	4.811	0.360	1.252	3.332	0.173
350	12.226	8.821	1.209	4.071	5.613	0.490	1.665	3.887	0.235
400	15.656	10.081	1.580	5.213	6.415	0.640	2.132	4.442	0.307
450	19.473	11.341	1.999	6.484	7.217	0.809	2.652	4.997	0.388
500	23.668	12.601	2.468	7.881	8.019	0.999	3.223	5.553	0.479
550	28.238	13.861	2.986	9.402	8.820	1.209	3.846	6.108	0.580
600	33.175	15.121	3.554	11.046	9.622	1.439	4.518	6.663	0.690
650	38.476	16.382	4.171	12.811	10.424	1.689	5.240	7.218	0.810
700	44.136	17.642	4.837	14.696	11.226	1.959	6.011	7.774	0.939
750	50.152	18.902	5.553	16.699	12.028	2.248	6.830	8.329	1.078
800	56.520	20.162	6.318	18.819	12.830	2.558	7.697	8.884	1.227
850	63.235	21.422	7.132	21.055	13.631	2.888	8.612	9.439	1.385
900	70.296	22.682	7.996	23.407	14.433	3.238	9.574	9.995	1.553
950	77.700	23.942	8.909	25.872	15.235	3.608	10.582	10.550	1.730
1000	85.443	25.202	9.872	28.450	16.037	3.997	11.637	11.105	1.917
1100				33.942	17.641	4.837	13.883	12.216	2.319
1200				39.877	19.244	5.756	16.311	13.326	2.760
1300				46.249	20.848	6.755	18.917	14.437	3.239
1400				53.053	22.452	7.835	21.700	15.547	3.757
1500				60.284	24.056	8.994	24.657	16.658	4.313
1600				67.938	25.659	10.233	27.788	17.768	4.907

Nota: Las velocidades en la succión no deben exceder los 8 pies por segundo.